Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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The length of the abstract should be approximately 150 words.

1 Introduction

The guidelines given below, including complete descriptions of the fonts, spacing, and related information for producing your manuscripts, are critical in order to produce the ICAD 2024 proceedings with a more uniform look. If you have any questions, please email them to the paper chairs.

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All manuscripts must be formatted for white A4 paper (8.27 in x 11.02 in). Please do **not** use US letter-size papers. All printed material, including text, illustrations, and charts, must be kept within a print area of 6.27 inches (169 mm) wide by 9.2 inches (233 mm) high. Do not write or print anything outside the print area. The top margin must be 1 inch (25 mm), except for the title page, and the left margin must be 0.75 inch (19 mm). All *text* must be in a two-column format. Columns are to be 3.29 inches (83.5 mm) wide, with a 0.31 inch (8 mm) space between them. Text must be fully justified.

3 Page Title Section

The paper title (on the first page) should begin 0.98 inches (25 mm) from the top edge of the page, centered, completely capitalized, and in Times New Roman 14-point, boldface type. The authors' name(s) and affiliation(s) appear below the title. Papers with multiple authors and affiliations may require a greater amount of space for this information.

4 Type-Style and Fonts

To achieve the best rendering in the proceedings, we strongly encourage you to use Times New Roman font. In addition, this will give the proceedings a more uniform look. Use a font size that is no smaller than nine point type throughout the paper, including figure captions.

In nine point type font, capital letters are 2 mm high. **If you use the smallest point size, there should be no more than 3.2 lines/cm (8 lines/inch) vertically.** This is a minimum spacing; 2.75 lines/cm (7 lines/inch) will make the paper much more readable. Larger type sizes require correspondingly larger vertical spacing. Please do not double-space your paper. True-Type 1 fonts are preferred.

The first paragraph in each section should not be indented, but all the following paragraphs within the section should be indented as these paragraphs demonstrate.

5 Major Headings

Major headings, as for this section, should appear in all capital letters, bold face, centered in the column, with one blank line before, and one blank line after. Use a period (".") after the heading number, not a colon.

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Subheadings should appear in sentence case and in boldface. They should start at the left margin on a separate line.

5.1.1 Sub-subheadings

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Illustrations must appear within the designated margins. They may span the two columns. If possible, position illustrations at the top of columns rather than in the middle or at the bottom. Caption and number every illustration. All halftone illustrations must be clear black and white prints. Colors may be used, but they should be selected so as to be readable when printed on a black and white-only printer.

Since there are many ways, often incompatible, of including images (e.g., with experimental results) in a LaTeX document, an example of how to do this is presented in Fig. 1.

figure=pics/fig1a-cropped.png,width=0.8
(a) Result 1

figure=pics/fig1b-cropped.pdf,height=4.0cm
(b) Result 2

figure=pics/fig1c-cropped.pdf,height=4.0cm
(c) Result 3

Gambar 1: Example of placing a figure with experimental results. Please use detailed figure captions that explain the figure well. The more modern way to include figures is to use *pdflatex* instead of *latex* as the compiler and then to use the \`includegraphics` command with pdf files.

8 Equations

Equations should be placed on separate lines and consecutively numbered with equation numbers in parentheses flush with the right margin, as illustrated in (1), which gives the homogeneous acoustic wave equation in Cartesian coordinates [?],

$$\nabla^2 p(x,y,z,t) - \frac{1}{c^2}\frac{\partial^2 p(x,y,z,t)}{\partial t^2} = 0, \tag{1}$$

where $p(x,y,z,t)$ is an infinitesimal variation of acoustic pressure from its equilibrium value at position (x,y,z) and time t, and where c denotes the speed of sound.

Symbols in your equation should be defined before the equation appears or immediately following it. Use (1), not Eq. (1) or equation (1), except at the beginning of a sentence: "Equation (1) is ..."

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10 REFERENCES

List and number all bibliographical references at the end of the paper. The references should be numbered in order of appearance in the document. When referring to them in the text, place the corresponding reference number in square brackets as shown at the end of this sentence [?], [?]. For LaTeX users, the use of the BibTeX style file IEEEtran.bst is recommended, as included in this template.

11 ACKNOWLEDGMENT

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